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Reg.No: .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBTS301 - MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A MCQ Answer all questions

Each question carries 1 mark

1.The regulatory protein that binds to the operator of the lac operon is the: **[U] / [CO2]**

a). Repressor

2. Which RNA carries amino acids to the ribosome during protein synthesis? **[K] / [CO1]**

c). tRNA

3. The central dogma describes the flow of information from **[K] / [CO1]**

c). DNA → RNA → Protein

4. RISC stands for: **[K] / [CO4]**

a). RNA-induced silencing complex

5. The process by which the ribosome moves along mRNA is called: **[K] / [CO1]**

b). Translocation

6. The promoter is a DNA sequence that primarily functions as: **[K] / [CO2]**

b). A site for initiation of transcription

7. The 5′ cap of eukaryotic mRNA contains: **[U] / [CO2]**

c). 7-methylguanosine

8. DNA polymerase synthesizes the new DNA strand in which direction? **[K] / [CO1]**

b). 5′ → 3′

9. Which is a stop codon? **[K] / [CO1]**

a) and c) UAA

10. Which radioactive isotope was used to label protein? **[K] / [CO1]**

c). ^{35}S

[1x10 = 10]

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

11. What is the role of topoisomerase? **[K] / [CO1]**

Topoisomerase relieves DNA supercoiling and tension during DNA replication and transcription

12. Define transcription. **[K] / [CO1]**

Transcription is the process of synthesizing RNA from a DNA template using RNA polymerase.

13. What is attenuation in the trp operon? **[U] / [CO2]**

Attenuation is a regulatory mechanism that terminates transcription prematurely in the trp operon when tryptophan levels are high.

14. Define tRNA. **[K] / [CO1]**

Transfer RNA (tRNA) is an RNA molecule that carries specific amino acids to the ribosome during protein synthesis.

15. List the major stages of translation. **[K] / [CO1]**

The three major stages of translation are initiation, elongation and termination.

16. What is photoreactivation? **[K] / [CO1]**

Photoreactivation is a DNA repair mechanism in which the enzyme photolyase uses visible or blue light energy to repair UV-induced pyrimidine dimers

17. What was the contribution of Erwin Chargaff to DNA structure? **[K] / [CO1]**

Erwin Chargaff proposed that in double-stranded DNA, the total amount of purines equals the total amount of pyrimidines. Thus, adenine equals thymine (A = T), and guanine equals cytosine (G = C).

18. What is a regulator gene? Give one example **[K] / [CO2]**

A regulator gene encodes a product that controls the expression of other genes. Example: the lacI gene of the lac operon.

19. What is conservative DNA replication? **[U] / [CO1]**

Conservative DNA replication is a model in which the parental DNA double helix remains intact, and an entirely new double helix is synthesized.

20. Why are aminoacyl-tRNA synthetases called "second genetic code readers"? [U] / [CO2]

They are called second genetic code readers because they recognize specific tRNAs and attach the correct amino acids to them, ensuring accurate translation of the genetic code.

21. What is siRNA? [K] / [CO4]

Small interfering RNA (siRNA) is a short double-stranded RNA molecule that silences specific genes by promoting the degradation of complementary mRNA.

22. What is gene therapy? [U] / [CO5]

Gene therapy is a technique used to treat or prevent diseases by introducing, replacing, modifying or silencing genetic material in a patient's cells.

[2x10 = 20]

Part C

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

23. Differentiate between cistron, muton and recon. [U] / [CO1]

Cistron (2 marks).
A cistron is a functional unit of genetic material. It represents a segment of DNA containing a sequence of nucleotides that controls a particular cellular function.

Important features

- **A cistron is a functional unit of a gene.**
- **A protein-coding cistron begins with a start codon and ends with a termination (stop) codon.**
- **It generally contains the information required to produce one polypeptide chain.**

Recon (1 mark).
A recon is the smallest unit of DNA capable of undergoing genetic recombination.

Features

- **It is considered a unit of recombination.**
- **Recombination can occur between different recons within a genetic region.**

Muton (1 mark)
A muton is the smallest unit of DNA capable of undergoing a mutation.

Features

- **It is considered a unit of mutation.**
- **A muton may consist of one or more nucleotide pairs.**

- **A change in the nucleotide sequence of a muton can result in a mutation.**

24. Describe the structure of DNA. [U] / [CO1]

DNA (Deoxyribonucleic acid) is a double-helical molecule proposed by Watson and Crick in 1953.

- **Double helix: DNA consists of two polynucleotide strands twisted around each other.**
- **Nucleotides are joined by 3′–5′ phosphodiester bonds between adjacent nucleotides, forming the sugar-phosphate backbone of each DNA strand.**
- **Antiparallel strands: One strand runs in the 5′ → 3′ direction and the other in the 3′ → 5′ direction.**
- **Nucleotide composition: Each nucleotide contains deoxyribose sugar, a phosphate group and a nitrogenous base (A, T, G or C).**
- **Complementary base pairing: Adenine pairs with thymine through two hydrogen bonds, while guanine pairs with cytosine through three hydrogen bonds.**
- **Dimensions: B-DNA has approximately 10 base pairs per turn, a pitch of 3.4 nm and a diameter of 2 nm.**
- **Major groove: The wider groove in the DNA double helix, where proteins can bind and recognize specific DNA sequences.**
- **Minor groove: The narrower groove in the DNA double helix, which also serves as a binding site for certain proteins and other molecules. (0.5 marks for each point.)**

(1 mark can be awarded for a properly labelled diagram, even if the explanation is not provided.)

25. Discuss the ethical, legal and social implications of the Human Genome Project. [U] / [CO5]

The Human Genome Project (HGP) was an international research initiative to sequence and map the entire human genome.

- **Ethical issues: Concerns regarding genetic privacy, informed consent and the use of genetic information.**
- **Legal issues: Genetic discrimination in employment and insurance.**
- **Social issues: Stigmatization of individuals or groups based on genetic characteristics.**
- **Benefits and concerns: Improved disease diagnosis and personalized medicine, but unequal access to genetic technologies may increase social inequalities. (1 mark for each point.)**

26. Differentiate between theta, rolling circle and D-loop replication. [U] / [CO1]

Feature	Theta replication	Rolling circle replication	D-loop replication
Structure	Forms a structure resembling the Greek letter theta (θ).	Produces a circular template with a displaced single strand.	Forms a displacement loop (D-loop).

Initiation	**Begins at an origin in circular DNA.**	**A nick is introduced in one strand of circular DNA.**	**Replication begins at a specific origin, commonly in mitochondrial DNA.**
Mechanism	**Replication proceeds from the origin, usually bidirectionally.**	**DNA synthesis proceeds around the circle, displacing the old strand.**	**A new strand displaces one parental strand during synthesis.**

(Structure and initiation points – 1 mark each) and (Mechanism – 2 marks.)

27. Describe the different methods of post translation modification of protein. [U] / [CO2]

Post-translational modifications (PTMs) are chemical or structural changes that occur in proteins after translation and help regulate their activity and stability.

Major types include:

- **Phosphorylation: Addition of phosphate groups, often regulating enzyme activity and cell signalling.**
- **Glycosylation: Addition of carbohydrate groups, important in protein folding and cell recognition.**
- **Acetylation: Addition of acetyl groups, which can regulate protein function and gene expression.**
- **Methylation: Addition of methyl groups, involved in regulating histones and other proteins.**
- **Ubiquitination: Attachment of ubiquitin, often marking proteins for degradation.**
- **Proteolytic cleavage: Removal of specific peptide segments to activate proteins, as in the conversion of proinsulin to insulin.**
- **Disulfide bond formation: Formation of bonds between cysteine residues that stabilize protein structure. (Explain any four methods – 4 marks.)**

28. Explain the A, P and E sites of ribosomes. [U] / [CO1]

The ribosome contains three major tRNA-binding sites involved in translation.

<u>A site (Aminoacyl site)</u>

Receives the incoming aminoacyl-tRNA carrying the next amino acid.

<u>P site (Peptidyl site)</u>

Holds the tRNA carrying the growing polypeptide chain. Peptide bond formation and chain transfer occur during elongation.

<u>E site (Exit site)</u>

Temporarily holds the deacylated tRNA before it leaves the ribosome.

(Introduction-1 mark, Explain the three different sites- 3 marks.)

29. Describe the major levels at which eukaryotic gene expression is regulated. [U] / [CO2]

Gene expression in eukaryotes is regulated at several levels to control the amount of protein production.

1. **Epigenetic and transcriptional regulation: DNA methylation, histone modifications and transcription factors regulate the accessibility of DNA and initiation of transcription.**
2. **Post-transcriptional regulation: RNA processing, including capping, splicing and polyadenylation, along with mRNA transport and stability, controls mature mRNA availability.**
3. **Translational regulation: Controls the efficiency with which mRNA is translated into protein.**
4. **Post-translational regulation: Protein modification, folding, transport and degradation regulate the activity and lifespan of proteins. (1 mark for each point.)**

[4x5 = 20]